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United States Patent	12408392
Kind Code	B2
Date of Patent	September 02, 2025
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Semiconductor device and method of forming the same

Abstract

The present disclosure relates to a semiconductor device and a method of forming the same. The semiconductor device includes: a substrate; a transistor on the substrate, which includes an active cylinder, the active cylinder includes a channel region, a source region and a drain distributed on opposite sides of the channel region, a first doped region located between the source region and the channel region, and a second doped region located between the drain region and the channel region, the first doped impurity region, the source region, the second doped impurity region and the drain region, all these regions include doped ions of the first type, and the doped concentration of the first impurity region is lower than that of the source region impurity concentration, the doped concentration of the second doped region is lower than the doped concentration of the drain region. The present disclosure reduces the band-to-band tunneling effect inside the transistor, thereby reducing the GIDL effect.

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Appl. No.:	17/908467
Filed (or PCT Filed):	July 20, 2022
PCT No.:	PCT/CN2022/106659
PCT Pub. No.:	WO2023/245788
PCT Pub. Date:	December 28, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240194731 A1	Jun. 13, 2024

Foreign Application Priority Data

CN

202210705481.0

Jun. 21, 2022

Publication Classification

Int. Cl.: H10D62/10 (20250101); H10B12/00 (20230101); H10D84/83 (20250101)

U.S. Cl.:

CPC H10D62/115 (20250101); H10B12/01 (20230201); H10B12/488 (20230201); H10D84/83 (20250101);

Field of Classification Search

CPC: H10D (62/115); H10D (84/83); H10D (62/10); H10D (62/124); H10B (12/01); H10B (12/488); H10B (12/00)

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Background/Summary

CROSS REFERENCES TO RELATED APPLICATIONS

(1) This application claims the benefit of priority of Chinese patent application with the application number CN 202210705481.0, entitled “A Semiconductor Device and Method of Forming the Same”, filed on Jun. 21, 2022, with the China National Intellectual Property Administration, the contents of which are incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to the technical field of semiconductor manufacturing, and in particular, to a semiconductor device and a method for forming the same.

BACKGROUND

(3) A Dynamic Random Access Memory (DRAM) is a semiconductor device commonly used in electronic equipment such as computers, and includes a plurality of memory cells, each of which usually includes a transistor and a capacitor. The gate of the transistor is electrically connected to the word line, the source is electrically connected to the bit line, and the drain is electrically connected to the capacitor. The word line voltage on the word line can control the opening and closing of the transistor, so that the memory can read through the bit line to get data information in the capacitor, or write the data information into the capacitor.

(4) A transistor of a semiconductor device such as the DRAM generally further includes a channel region, a source regions and a drain region located on opposite sides of the channel region and adjacent to the channel region. As the size of DRAMs continues to shrink, the BTBT (Band to Band Tunneling) effect inside the transistor is enhanced, resulting in an enhanced GIDL (Gate-Induced Drain Leakage) effect inside the semiconductor device. This eventually leads to the degradation of the electrical properties of the semiconductor device.

(5) Therefore, how to reduce the GIDL effect inside the semiconductor device and improve the performance of the semiconductor device is an urgent technical problem to solved at the present time.

SUMMARY

(6) Some embodiments of the present disclosure provide a semiconductor device and a method for forming the same, which are used to reduce the GIDL effect inside the semiconductor device, thereby improving the electrical performance of the semiconductor device and improving the yield of the semiconductor device.

(7) According to some embodiments, the present disclosure provides a semiconductor device comprising: a substrate; and a transistor located on the substrate and including an active cylinder, the active cylinder includes a channel region, a source region and a drain region distributed on opposite sides of the channel region, a first doped region between the source region and the channel regions, and a second doped region between the drain region and the channel region; the first doped region, the source region, the second doped region and the drain region contain doping ions of the first type; the doped concentration of the first doped region is lower than the doped concentration of the source region, and the doped concentration of the second doped region is less than the doped concentration of the drain region.

(8) In some embodiments, the active cylinders extends along a first direction, the first direction being a direction parallel to the top surface of the substrate. The semiconductor device further includes: a first isolation layer, located on the substrate, wherein the first isolation layer is distributed around the periphery of the first doped region, and the active cylinder penetrates the first isolation layer along the first direction; and a second isolation layer, located on the substrate, wherein the second isolation layer is distributed around the outer periphery of the second doped region, and the active cylinder penetrates the second isolation layer along the first direction.

(9) In some embodiments, a plurality of the active cylinders are arranged at intervals along a second direction, the second direction is a direction parallel to the top surface of the substrate, and the first direction and the second direction intersect;

(10) The first isolation layer continuously surrounds the outer periphery of the plurality of first doped regions arranged at intervals along the second direction; and the second isolation layer continuously surrounds the outer periphery of the plurality of second doped regions arranged at intervals along the second direction.

(11) In some embodiments, it also includes: a first air gap, located between the adjacent first doped regions; and a second air gap, located between the adjacent second doped regions.

(12) In some embodiments, the active cylinders extend along a first direction, the first direction being a direction parallel to the top surface of the substrate. The semiconductor device further includes: a first isolation layer located on the substrate, and the first isolation layer covers the first doped region, the active cylinder penetrates the first isolation layer along the first direction; and a second isolation layer, located on the substrate, and the second isolation layer covers the second doped region, and the active cylinder penetrates the second isolation layer along the first direction.

(13) In some embodiments, a plurality of the active cylinders are arranged at intervals along a second direction, the second direction is a direction parallel to the top surface of the substrate, and the first direction and the second direction intersect; the first isolation layer continuously covers a plurality of the first doped regions spaced along the second direction; and

(14) The second isolation layer continuously covers the plurality of the second doped regions spaced along the second direction.

(15) In some embodiments, the channel region includes dopant ions of the first type; or,

(16) The channel region includes dopant ions of a second type, and the dopant ions of the first type and the dopant ions of the second type are of opposite conductivity.

(17) In some embodiments, the active cylinders extend in a first direction, the first direction being a direction parallel to a top surface of the substrate.

(18) The length of the first doped region along the first direction is more than twice the thickness of the active cylinder along a third direction, and the third direction is a direction perpendicular to the top surface of the substrate.

(19) The length of the second doped region along the first direction is more than twice the thickness of the active cylinder along the third direction.

(20) In some embodiments, the active cylinders extend in a first direction, the first direction being a direction parallel to a top surface of the substrate;

(21) The length of the first doped region along the first direction is greater than or equal to the length of the source region along the first direction.

(22) The length of the second doped region along the first direction is greater than or equal to the length of the drain region along the first direction.

(23) In some embodiments, the active cylinders extend along a first direction, and a plurality of the active cylinders are arranged in arrays along the second direction and the third direction, both of the first direction and the second direction are parallel to the top surface of the substrate, and the first direction intersects the second direction, and the third direction is a direction perpendicular to the top surface of the substrate. The semiconductor device further includes: word lines extending along the second direction and continuously covering the channel regions in a plurality of the active cylinders spaced along the second direction; and a plurality of the word lines arranged at intervals along the third direction, and among the two adjacent word lines along the third direction, one of the word lines closer to the substrate protrudes from another word lines along the second direction.

(24) According to other embodiments, the present disclosure also provides a method for forming a semiconductor device, comprising the following steps: provide a substrate; forming a transistor on the substrate, the transistor includes an active cylinder, the active cylinder includes a channel region, a source region and a drain region distributed on opposite sides of the channel region, and the source region; a first doped region disposed between the source region and the channel region, and a second doped region between disposed between the drain region and the channel region; the

first doped region, the source region, the second doped region and the drain region all include doping ions of the first type; and the doped concentration of the first doped region is lower than the doped concentration of the source region, and the doped concentration of the second doped region is lower than the doped concentration of the drain region.

(25) In some embodiments, the specific steps of forming transistors on the substrate include: forming semiconductor cylinders over the substrate; defining an initial channel region in the semiconductor cylinder, initial source regions and initial drain regions distributed on opposite sides of the initial channel region, and between the initial source region and the initial channel region an initial first doped region, and an initial second doped region between the initial drain region and the initial channel region; and implanting a first type doping ions into the initial source region, the initial drain region, the initial first doped region and the initial second doped region to form the source region, the first doped region a doped region, the drain region and the second doped region.

(26) In some embodiments, an initial channel region, an initial source region and an initial drain region distributed on opposite sides of the initial channel region, and the initial source region and the initial drain region are defined in the semiconductor cylinder. The specific steps of the initial first doped region between the initial channel regions and the initial second doped region between the initial drain region and the initial channel region include: defining an initial channel region in the semiconductor cylinder, an initial source region and an initial drain region distributed on opposite sides of the initial channel region along a first direction, and an initial source region along the first direction, an initial first doped region between the initial channel region and the initial channel region, and an initial second doped region located between the initial drain region and the initial channel region along the first direction, so the length of the initial first doped region along the first direction is more than twice the thickness of the semiconductor cylinder along the third direction; and the length of the initial second doped region along the first direction is more than twice the thickness of the semiconductor cylinder along third direction, wherein the first direction is a direction parallel to the top surface of the substrate, and the third direction is a direction perpendicular to the top surface of the substrate.

(27) In some embodiments, the specific steps of forming semiconductor cylinders over the substrate include: forming a stacked layer on the substrate, the stacked layer including a semiconductor layer and a sacrificial layer alternately stacked along a third direction, the third direction being a direction perpendicular to the top surface of the substrate; etching the stacked layer to form a first trench penetrating the stacked layer along the third direction, a plurality of the first trenches are arranged at intervals along the second direction, and the first trench connects the semiconductor layer is separated into a plurality of the semiconductor cylinders spaced along the second direction, the semiconductor cylinders extend along the first direction, and both the first direction and the second direction are parallel to the top surface of the substrate, and the first direction intersects the second direction.

(28) In some embodiments, before implanting the first type dopant ions into the initial source region, the initial drain region, the initial first doped region and the initial second doped region, the method further includes the following: step: removing a part of the sacrificial layer from the stacked layer to form second trenches and third trenches, the second trenches expose the semiconductor cylinders between the initial first doped region and the initial channel region and the semiconductor cylinders between the initial first doped region and the initial source region, the third trenches expose the semiconductor cylinder between the initial second doped region and the initial channel region and the semiconductor cylinder between the initial second doped region and the initial drain region; depositing a first dielectric material in the second trenches and the third trenches, forming a first isolation layer in the second trenches, and forming a second isolation layer in the third trenches at the same time.

(29) In some embodiments, before implanting the first type dopant ions into the initial source region, the initial drain region, the initial first doped region and the initial second doped region, the

method further includes the following steps: removing the sacrificial layer in the stacked layer to expose the fourth trenches of the semiconductor cylinder; and filling the fourth trenches with a second dielectric material to form an interlayer isolation layer.

(30) In some embodiments, the specific steps of implanting the first type dopant ions into the initial source region, the initial drain region, the initial first doped region and the initial second doped region include: implanting first type dopant ions into the initial source region and the initial drain region to form the source region and the drain region; and implanting the first type of doping ions into the initial first doped region and the initial second doped region to form the first doped region and the second doped region.

(31) In some embodiments, it also includes: Implanting the first type dopant ions into the initial channel region to form the channel region; or, implanting the second type of dopant ions into the initial channel region to form the channel region, wherein the first type of dopant ions and the second type of dopant ions have opposite conductivity types.

(32) In some embodiments, after forming the source region, the first doped region, the drain region and the second doped region, it further includes the following steps: removing the interlayer isolation layer from between the adjacent first doped regions and simultaneously removing the interlayer isolation layer from between the adjacent second doped regions to form adjacent first doped regions, wherein a first air gap is formed between the adjacent first doped regions, and a second air gap is formed between the adjacent second doped regions simultaneously.

(33) In some embodiments, after forming the first air gap between the adjacent first doped regions and simultaneously forming the second air gap between the adjacent second doped regions, the method further comprises the following step: depositing the first dielectric material in the first air gap and the second air gap.

(34) Some embodiments of the present disclosure provide a semiconductor device and a method for forming the same, by arranging a first doped region between a channel region and a source region, and arranging a second doped region between the channel region and the drain region, and the ion doping type of the first doped region is the same as that of the source region, the ion doping type of the second doped region is the same as that of the drain region, and the doped concentration of the first doped region is lower than that of the source region and the second doped region. The doped concentration of the impurity region is lower than that of the drain region, so as to reduce the band to band tunneling effect inside the transistor, thereby reducing the GIDL effect. Furthermore, some embodiments of the present disclosure form a first isolation layer surrounding the first doped region and a second isolation layer surrounding the second doped region. On the one hand, the first doped region and the second doped region affect other components in the semiconductor device; on the other hand, it also helps to further reduce the band-band tunneling effect inside the transistor, thereby further reducing the GIDL effect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic top-view diagram of a semiconductor device structure in an embodiment of the present disclosure;

(2) FIG. 2 is a schematic top-view diagram of a semiconductor device structure in another embodiment of the present disclosure;

(3) FIG. 3 shows a method of forming a semiconductor device according to an embodiment of the present disclosure; and

(4) FIGS. 4-11 are schematic diagrams at major steps in the main process of forming a semiconductor device according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(5) The specific embodiments of the semiconductor device and the method thereof provided by the present disclosure will be described in detail below with reference to the accompanying drawings.

(6) This embodiment is provided by FIG. 1, is a schematic top-view diagram of the semiconductor device structure in an embodiment of the present disclosure. As shown in FIG. 1, the semiconductor device includes: a substrate; the transistor is located on the substrate, and includes an active cylinder, the active cylinder includes a channel region 10, a source region 11 and a drain region 12 disposed on opposite sides of the channel region 10, located on the first doped region 13 between the source region 11 and the channel region 10, and a second doped region 14 between the drain region 12 and the channel region 10. The first doped region 13, the source region 11, the second doped region 14 and the drain region 12 all include doping ions of the first type, and the doped concentration of the first doped region 13 is less than the doped concentration of the source region 11, and the doped concentration of the second doped region 14 is lower than the doped concentration of the drain region 12.

(7) Specifically, the substrate may be, but not limited to, a silicon substrate, and this specific embodiment is described by taking the substrate being a silicon substrate as an example. In other embodiments, the substrate may also be a semiconductor substrate such as gallium nitride, gallium arsenide, gallium carbide, silicon carbide, or SOI. The substrate is used to support device structures thereon. The semiconductor device may be, but is not limited to, a DRAM, and this specific embodiment is described by taking the semiconductor device being a DRAM as an example, where the substrate includes a memory array having a plurality of memory cells arranged in an array, the memory cells include transistors, and capacitors 22 electrically connected to the transistors. The active cylinder in the transistor includes the channel region 10, the source region 11, the drain region 12, the first doped region 13 and the second doped region 14, wherein, the channel region 10 and the source region 11 are distributed on opposite sides of the first doped region 13 along the extending direction of the active cylinder, and the channel region 10 and the drain region 12 are distributed on opposite sides of the second doped region 14 along the extending direction of the active cylinders. The semiconductor device further includes support pillars 17 distributed around the periphery of the memory array and penetrating the memory array in a direction perpendicular to the top surface of the substrate. The support column 17 is used to support the storage array and to isolate the storage array from other peripheral components. In one embodiment, the material of the support pillar 17 may be a nitride material (e.g., silicon nitride). A plurality of in the present embodiment refers to two or more.

(8) The types of doping ions in the first doped region 13, the source region 11, the second doped region 14 and the drain region 12 are of the same type (all are the first type doping ions)), that is, the first doped region 13, the source region 11, the second doped region 14 and the drain region 12 are doped with the same type of ions. The doped concentration of the first type dopant ions in the first doped region 13 is lower than the doped concentration of the first type doping ions in the source region 11, and the second doped region 14. The doped concentration of the first type dopant ions in the drain region 12 is smaller than the doped concentration of the first type dopant ions in the drain region 12, that is, the first doped region 13 and the second doping ion region 14 is lightly doped, and the source region 11 and the drain region 12 are heavily doped. By arranging the first doped region 13 lightly doped with the same type as the source region 11, between the source region 11 and the channel region 10, and arranging the second doped region 14 lightly doped with the same type as the drain region 12, between the drain region 12 the channel regions 10, the band to band tunneling effect inside the transistor will be reduced, thereby reducing the GIDL effect.

(9) In some embodiments, the active cylinders extend along a first direction D1, the first direction D1 being a direction parallel to the top surface of the substrate. The semiconductor device further includes: the first isolation layer 1, located on the substrate, and the first isolation layer 15 is distributed around the outer circumference of the first doped region 13, and the active cylinder penetrates the first direction D1 along the first direction D1, an isolation layer 15; and the second

isolation layer **16**, located on the substrate, and the second isolation layer ring **16** is disposed around the outer periphery of the second doped region **14**, and the active cylinder penetrates through the second isolation layer **16** in the first direction **D1**.

(10) In some embodiments, a plurality of the active cylinders are arranged at intervals along a second direction **D2**, the second direction **D2** is a direction parallel to the top surface of the substrate, and the first direction **D1** and the second direction **D2** intersect.

(11) The first isolation layer **15** continuously surrounds the outer periphery of the plurality of first doped regions **13** arranged at intervals along the second direction **D2**.

(12) The second isolation layer **16** continuously surrounds the outer periphery of the plurality of second doped regions **14** arranged at intervals along the second direction **D2**.

(13) In some embodiments, it also includes: a first air gap, located between the adjacent first doped regions **13**; a second air gap, located between the adjacent second doped regions **14**.

(14) Specifically, as shown in FIG. **1**, the active cylinders extend along the first direction **D1**, and a plurality of the active cylinders extend along the second direction **D2** and perpendicular to the top surface of the substrate. The directions are arranged in an array. The top surface of the substrate refers to the surface of the substrate facing the active cylinder. Correspondingly, the plurality of first doped regions **13** are arranged in an array along the second direction **D2** and the direction perpendicular to the top surface of the substrate, and the plurality of second doped regions **14** are also arranged along the second direction **D2**, which is the direction perpendicular to the top surface of the substrate are arranged in an array. The first isolation layer **15** is in the shape of a surrounding frame, extends in a direction perpendicular to the top surface of the substrate, and is continuously distributed around the periphery of the plurality of first doped regions **13** arranged in an array. The second isolation layer **16** is also in the shape of a surrounding frame, extends in a direction perpendicular to the top surface of the substrate, and is continuously distributed around the periphery of the plurality of second doped regions **14** arranged in an array. The first doped region **13** is isolated from signal lines such as word lines in the semiconductor device by the first isolation layer **15**, and the second doped region **14** is isolated from the second doped region **14** by the second isolation layer **16** signal lines such as word lines in the semiconductor device, thereby helping to further reduce the GIDL effect. In one embodiment, the materials of the first isolation layer **15** and the second isolation layer **16** are the same, for example, both are insulating dielectric materials such as nitride (e.g., silicon nitride). The first air gap is formed between the first doped regions **13** surrounded by the first isolation layer **15**, so that the low dielectric constant of air is used to reduce the gap between the adjacent first doped regions **13** interference. Correspondingly, the second air gap is formed between the second doped regions **14** surrounded by the second isolation layer **16**, so that the low dielectric constant of air is used to reduce the adjacent second doped regions **14** interference between.

(15) In one embodiment, the first isolation layer **15** and the second isolation layer **16** can be formed simultaneously with the support pillars **17**, thereby simplifying the manufacturing process of the semiconductor device and reducing the manufacturing cost of the semiconductor device.

(16) In other embodiments, the active cylinders extend along a first direction **D1**, and the first direction **D1** is a direction parallel to the top surface of the substrate; the semiconductor device further includes: The first isolation layer **15** is located on the substrate, and the first isolation layer **15** covers the first doped region **13**, and the active cylinder penetrates the first isolation along the first direction **D1** layer **15**; The second isolation layer **16** is located on the substrate, and the second isolation layer **16** covers the second doped region **14**, and the active cylinder penetrates the second isolation layer **16** along the first direction **D1**

(17) In other embodiments, a plurality of the active cylinders are arranged at intervals along a second direction **D2**, the second direction **D2** is a direction parallel to the top surface of the substrate, and the first direction **D1** and the second direction **D2** intersect.

(18) The first isolation layer **15** continuously covers a plurality of the first doped regions **13** spaced

along the second direction D2.

(19) The second isolation layer **16** continuously covers the plurality of the second doped regions **14** arranged at intervals along the second direction D2.

(20) Specifically, as shown in FIG. 2, the active cylinders extend along the first direction D1, and a plurality of the active cylinders extend along the second direction D2 and perpendicular to the top surface of the substrate. The directions are arranged in an array. The top surface of the substrate refers to the surface of the substrate facing the active cylinder. Correspondingly, the plurality of first doped regions **13** are arranged in an array along the second direction D2 and the direction perpendicular to the top surface of the substrate, and the plurality of second doped regions **14** are also arranged along the second direction D2 and the direction perpendicular to the top surface of the substrate are arranged in an array. The first isolation layer **15** continuously covers a plurality of the first doped regions **13** arranged in an array, and fills the gaps between the adjacent first doped regions **13**. The second isolation layer **16** continuously covers the plurality of second doped regions **14** arranged in an array, and fills the gaps between adjacent second doped regions **14**, thereby further reducing the band-to-band tunneling effect.

(21) In order to meet the requirements of various applications of the semiconductor device, in some embodiments, the channel region **10** includes the first type dopant ions; or,

(22) The channel region **10** includes dopant ions of a second type, and the dopant ions of the first type are of opposite conductivity types to the dopant ions of the second type.

(23) For example, the channel region **10**, the first doped region **13** and the second doped region **14** are all lightly doped with N-type ions, the source region **11** and the drain region **12** Both are heavily doped with N-type ions. Alternatively, the channel region **10**, the first doped region **13** and the second doped region **14** are all lightly doped with P-type ions, and the source region **11** and the drain region **12** are all lightly doped with P-type ions. P-type ions are heavily doped. Alternatively, the channel region **10** is lightly doped with N-type ions, the first doped region **13** and the second doped region **14** are both lightly doped with P-type ions, the source region **11** and all The drain region **12** is heavily doped with P-type ions. Alternatively, the channel region **10** is lightly doped with P-type ions, the first doped region **13** and the second doped region **14** are both lightly doped with N-type ions, and the source region **11** and all the drain region **12** are heavily doped with N-type ions.

(24) In some embodiments, the active cylinders extend along a first direction D1, the first direction D1 being a direction parallel to the top surface of the substrate.

(25) The length of the first doped region **13** along the first direction D1 is more than twice the thickness of the active cylinder along the third direction, and the third direction is perpendicular to the top surface of the substrate. direction.

(26) The length of the second doped region **14** along the first direction D1 is more than twice the thickness of the active cylinder along the third direction.

(27) For example, the thickness of the active cylinder along the third direction (i.e., the diameter of the active cylinder) may be greater than or equal to 5 nm, the first doped region **13** and the second doped region **14** The length along the first direction D1 may be greater than or equal to 10 nm. In this specific embodiment, the lengths of the first doped regions **13** and the second doped regions **14** along the first direction D1 are both set to be more than twice the thickness of the active cylinders along the third direction, which can better reduce or even eliminate the band-band tunneling effect, thereby reducing and setting the GIDL effect completely.

(28) In some embodiments, the active cylinders extend along a first direction D1, the first direction D1 being a direction parallel to the top surface of the substrate.

(29) The length of the first doped region **13** along the first direction D1 is greater than or equal to the length of the source region **11** along the first direction D1.

(30) The length of the second doped region **14** along the first direction D1 is greater than or equal to the length of the drain region **12** along the first direction D1.

(31) Specifically, the longer the length of the first doped region **13** and the length of the second doped region **14** are, the more significant the effect of reducing GIDL is. In order not to increase the size of the semiconductor device, along the first direction **D1**, the length of the first doped region **13** may be greater than the length of the source region **11**, and the length of the second doped region **14** is greater than the length of the drain region **12**.

(32) In some embodiments, the active cylinders extend along a first direction **D1**, a plurality of the active cylinders are arranged in an array along a second direction **D2** and a third direction, the first direction **D1** and the second direction **D2** are all directions parallel to the top surface of the substrate, and the first direction **D1** and the second direction **D2** intersect, and the third direction is a direction perpendicular to the top surface of the substrate. The semiconductor device further includes: word lines **18**, extending along the second direction **D2**, and continuously covering the channel regions **10** in the plurality of the active cylinders spaced along the second direction **D2**; and multiple word lines **18** are arranged at intervals along the third direction, and among the two adjacent word lines **18** along the third direction, the one word line **18** closer to the substrate protrudes above the other word line **18** along the second direction **D2**.

(33) Specifically, as shown in FIGS. **1** and **2**, the capacitor **22** is electrically connected to the drain region **12** in the transistor. The semiconductor device further includes bit lines **19** extending along the third direction, each of the bit lines **19** is in electrical contact with a plurality of the source regions **11** arranged at intervals along the third direction, and a plurality of the bit lines **19** are electrically connected to each other. The bit lines **19** are arranged at intervals along the second direction **D2**. The top surface of the bit line **19** is further provided with a bit line plug **21** for electrically connecting the bit line **19** and a peripheral control circuit. The semiconductor device further includes a plurality of the word lines **18** extending along the second direction **D2**, the plurality of the word lines **18** are spaced along the third direction, and each of the word lines **18** continuously covers a plurality of the channel regions **10** arranged at intervals alternatively along the second direction **D2** are covered. The ends of the word lines **18** extend out of the memory array to form a stepped structure, so that the word line plugs **20** are used to electrically connect the memory arrays. Word lines **18** and peripheral control circuits. The step-like structure refers to that, among the two adjacent word lines **18** along the third direction, one word line **18** closer to the substrate protrudes from the other along the second direction **D2** the word lines **18**.

(34) This specific embodiment also provides a method for forming a semiconductor device in FIG. **3**, according to an embodiment of the present disclosure; and

(35) FIGS. **4-11** are schematic diagrams at major steps in the main process of forming a semiconductor device according to an embodiment of the present disclosure.

(36) FIG. **4** is a schematic top view structure of the semiconductor device formed by this specific embodiment, and FIGS. **5-11** are the a-a position and the b-b position in FIG. **4** during the formation process of the semiconductor device, c-c position, d-d position and e-e position of the main process cross-sectional schematic diagram, to clearly show the formation process of the semiconductor device. The structure of the semiconductor device formed by this specific embodiment can be referred to FIG. **1** and FIG. **2**. The semiconductor device formed by this embodiment may be, but is not limited to, a DRAM. As shown in FIGS. **5-11**, the method for forming the semiconductor device includes the following steps:

(37) In step **S531**, a substrate **50** is provided, as shown in FIG. **5**.

(38) Specifically, the substrate **50** may be, but is not limited to, a silicon substrate, and this specific embodiment is described by taking the substrate **50** as a silicon substrate as an example. In other embodiments, the substrate **50** may also be a semiconductor substrate such as gallium nitride, gallium arsenide, gallium carbide, silicon carbide, or SOI. The substrate **50** is used to support device structures thereon.

(39) Step **S532**, forming a transistor on the substrate, the transistor includes an active cylinder, and

the active cylinder includes a channel region **10**, a source region **11** and a drain distributed on opposite sides of the channel region **10** and region **12**, a first doped region **13** between the source region **11** and the channel region **10**, and a second doped region between the drain region **12** and the channel region **10**. The first doped region **13**, the source region **11**, the second doped region **14** and the drain region **12** all include doping ions of the first type, and the doped concentration of the first doped region **13** is lower than the doped concentration of the source region **11**, and the doped concentration of the second doped region **14** is lower than that of the drain region **12**, as shown in FIG. **11** and FIG. **4**.

(40) In some embodiments, the specific steps of forming transistors on the substrate **50** include: forming semiconductor cylinders **76** on the substrate **50**, as shown in FIG. **6**; forming an initial channel region **90**, an initial source region **91** and an initial drain region **93** distributed on opposite sides of the initial channel region **90**, and an initial source region **91** and the initial source region **93** are defined in the semiconductor cylinder **76**, an initial first doped region **92** between the initial channel regions **90**, and an initial second doped region **94** located between the initial drain region **93** and the initial channel region **90**, as shown in FIG. **9**; Implanting first type dopant ions into the initial source region **91**, the initial drain region **93**, the initial first doped region **92** and the initial second doped region **94** to form the source region **11**. The first doped region **13**, the drain region **12** and the second doped region **14**.

(41) In some embodiments, an initial channel region **90**, an initial source region **91** and an initial drain region **93** distributed on opposite sides of the initial channel region **90** are defined in the semiconductor cylinder **76**, located in the initial source. The initial first doped region **92** between the source region **91** and the initial channel region **90** and the initial second doped region **94** between the initial drain region **93** and the initial channel region **90**. Specific steps include: an initial channel region **90** is defined in the semiconductor cylinder **76**, initial source regions **91** and **93** distributed on opposite sides of the initial channel region **90** along the first direction **D1**, along the first direction **D1** is located in the initial first doped region **92** between the initial source region **91** and the initial channel region **90**, and is located between the initial drain region **93** and the initial channel along the first direction **D1**. The initial second doped region **94** between the regions **90**, the length of the initial first doped region **92** along the first direction **D1** is more than twice the thickness of the semiconductor cylinder **76** along the third direction **D3**, so the length of the initial second doped region **94** along the first direction **D1** is more than twice the thickness of the semiconductor cylinder **76** along the third direction **D3**, and the first direction **D1** is parallel to the top surface of the substrate **50**, the third direction **D3** is a direction perpendicular to the top surface of the substrate **50**.

(42) In some embodiments, the specific steps of forming the semiconductor cylinders **76** over the substrate **50** include: forming a stacked layer on the substrate **50**, and the stacked layer includes a semiconductor layer **52** and a sacrificial layer **51** alternately stacked along a third direction **D3**, the third direction **D3** is perpendicular to the top surface of the substrate **50** direction; etching the stacked layers to form first trenches **60** penetrating the stacked layers along the third direction **D3**. A plurality of the first trenches **60** are arranged at intervals along the second direction **D2**. The trench **60** separates the semiconductor layer **52** into a plurality of the semiconductor cylinders **76** spaced along the second direction **D2**, the semiconductor cylinders **76** extending along the first direction **D1**, the first direction **D1** and the second directions **D2** are all directions parallel to the top surface of the substrate **50**, and the first direction **D1** and the second direction **D2** intersect.

(43) Specifically, epitaxial growth can be used to alternately form the semiconductor layer **52** and the sacrificial layer **51** on the top surface of the substrate **50** along the third direction **D3** to form all the superlattice stack structures. The stacked layers are shown in FIG. **5**. The more the semiconductor layers **52** and the sacrificial layers **51** are alternately stacked, the greater the storage capacity of the formed semiconductor device. In one embodiment, the material of the semiconductor layer **52** may be Si, and the material of the sacrificial layer **51** may be SiGe. After

that, a first dielectric layer **53** is formed on the top surface of the stacked layer, and the stacked layer and the first dielectric layer **53** are etched by a photolithography process to form a plurality of penetrating layers along the third direction **D3**. The first dielectric layer **53** and the first trench **60** of the stacked layer. The first trench **60** separates the semiconductor layer **52** into a plurality of the semiconductor cylinders **76** spaced along the second direction **D2**, and the semiconductor cylinders **76** extend along the first direction **D1**, as shown in FIG. 6.

(44) In some embodiments, first type dopant ions are implanted before the initial source region **91**, the initial drain region **93**, the initial first doped region **92** and the initial second doped region **94**, also includes the following steps: removing a part of the sacrificial layer **51** from the stacked layers to form a second trench **73** and a third trench **74**, and the second trench **73** exposes the initial first doped region **92** and the initial channel. The semiconductor cylinders **76** between regions **90**, and the semiconductor cylinders **76** between the initial first doped regions **92** and the initial source regions **91**, the third trenches **74** expose the initial first semiconductor cylinder **76** between the second doped region **94** and the initial channel region **90**, and the semiconductor cylinder **76** between the initial second doped region **94** and the initial drain region **93**, such as shown in FIG. 7; and depositing a first dielectric material on the second trench **73** and the third trench **74**, forming the first isolation layer **15** in the second trench **73** and simultaneously forming the third trench **74**. The second isolation layer **16** is shown in FIGS. 8 and 4.

(45) Specifically, after the first trench **60** is formed, an insulating dielectric material such as oxide (eg, silicon dioxide) may be deposited in the first trench **60** to form the first trench **60** filled with A fill layer **70** covering the top surface of the stacked layers is shown in FIG. 7. After that, a photolithography process may be used to remove part of the sacrificial layer **51** to form the second trench **73**, the third trench **74**, and the support hole **71** exposing part of the semiconductor cylinder **76**. The second trench **73** is annular, and the projection of the second trench **73** on the top surface of the substrate **50** surrounds a plurality of the initial first doped regions **92** on the substrate. The perimeter distribution of the projection on the top surface of **50**. The third trench **74** is also annular, and the projection of the third trench **74** on the top surface of the substrate **50** surrounds a plurality of the initial second doped regions **94** on the substrate **50**. The peripheral distribution of the projection on the top surface of. At the same time, a first dielectric material such as nitride (e.g., silicon nitride) is filled in the second trench **73**, the third trench **74** and the support hole **71**, and the first isolation layer **15** is formed at the same time, the second isolation layer **16** and the support column **17**, as described in FIG. 4 and FIG. 8.

(46) In some embodiments, first type dopant ions are implanted before the initial source region **91**, the initial drain region **93**, the initial first doped region **92** and the initial second doped region **94**, also includes the following steps: removing the sacrificial layer **51** from the stacked layers to expose the fourth trench of the semiconductor cylinder **76**, as shown in FIG. 9; and filled the second dielectric material in the fourth trench to form an interlayer isolation layer **100**, as shown in FIG. 10.

(47) Specifically, a wet etching process may be used to remove all the sacrificial layers **51** in the stacked layers, thereby forming fourth trenches exposing the semiconductor cylinders **76**. After that, a second dielectric material such as oxide (e.g., silicon dioxide) is deposited in the fourth trench to form the interlayer isolation layer **100**, as shown in FIG. 10.

(48) In some embodiments, first type dopant ions are implanted into the initial source region **91**, the initial drain region **93**, the initial first doped region **92** and the initial second doped region **94**. Specific steps include: implanting first type dopant ions into the initial source region **91** and the initial drain region **93** to form the source region **11** and the drain region **12**; implanting the first type of doping ions into the initial first doped region **92** and the initial second doped region **94** to form the first doped region **13** and the second doped region **14**.

(49) In some embodiments, the method of forming the semiconductor device further comprises: implanting first type dopant ions into the initial channel region **90** to form the channel region **10**;

or, implanting the second type of dopant ions into the initial channel region **90** to form the channel region **10**, and the first type of dopant ions and the second type of dopant ions have opposite conductivity types.

(50) Specifically, multiple photomasks and multiple doping processes may be used to provide the initial source region **91**, the initial drain region **93**, the initial first doped region **92** and the initial second doped region **91**. The doped region **94** and the initial channel region **90** are implanted with doping ions respectively, and the doping of different types and/or different concentrations has been completed in each region. Wherein, the first type of doping ions are implanted into the initial source region **91**, the initial drain region **93**, the initial first doped region **92** and the initial second doped region **94**, and the specific method for doping the first type dopant ions or the second type dopant ions in the initial channel region **90** may be vapor diffusion, plasma doping or ion implantation.

(51) In some embodiments, after forming the source region **11**, the first doped region **13**, the drain region **12** and the second doped region **14**, the following steps are further included: removing the interlayer isolation layer **100** between the adjacent first doped regions **13** and removing the interlayer isolation layer **100** between the adjacent second doped regions **14** simultaneously, to form the first air gap between the adjacent first doped regions **13** and the second air gap between the adjacent second doped regions **14** simultaneously.

(52) Specifically, the first air gap is formed between the first doped regions **13** surrounded by the first isolation layer **15**, so that the low dielectric constant of air is used to reduce the adjacent first doped regions interference between zones **13**. Correspondingly, the second air gap is formed between the second doped regions **14** surrounded by the second isolation layer **16**, so that the low dielectric constant of air is used to reduce the adjacent second doped regions **14** interference between.

(53) In some embodiments, after forming first air gap between adjacent first doped regions **13** and forming second air gap between adjacent second doped regions **14** at the same time, further includes the following step: depositing the first dielectric material in the first air gap and the second air gap.

(54) Specifically, the first dielectric material continuously covers a plurality of the first doped regions **13** arranged in an array, and fills the gaps between the adjacent first doped regions **13**. Two dielectric materials continuously cover the plurality of second doped regions **14** arranged in an array, and fill the gaps between adjacent second doped regions **14**, thereby further reducing the band-to-band tunneling effect.

(55) After forming the source region **11**, the first doped region **13**, the drain region **12** and the second doped region **14**, a capacitor **22** is electrically connected to the drain region **12**, thus a capacitor **22** is formed. The source region **11** is electrically connected to the bit line **19**, the word line **18** covering the channel region **10**, the bit line plug **21** located on the top surface of the bit line **19**, and the top surface of the word line **18**. word line plugs **20** to obtain the structure shown in FIG. 4. A plurality of the word lines **18** are arranged at intervals along the third direction, and each of the word lines **18** continuously covers or wraps a plurality of the channel regions **10** arranged at intervals along the second direction D2. The ends of the plurality of word lines **18** together form a stepped structure, so that the word lines **18** and the peripheral control circuit are electrically connected through the word line plugs **20**.

(56) Some embodiments of the present invention provide a semiconductor device and a method for forming the same, by arranging a first doped region between a channel region and a source region, and arranging a second doped region between the channel region and the drain region. The ion doping type of the first doped region is the same as that of the source region, the ion doping type of the second doped region is the same as that of the drain region, and the doped concentration of the first doped region is lower than that of the source region and the second doped region. The doped concentration of the second doped region is lower than that of the drain region, so as to reduce the

band-band tunneling effect inside the transistor, thereby reducing the GIDL effect. Furthermore, some embodiments of the present disclosure form a first isolation layer surrounding the first doped region and a second isolation layer surrounding the second doped region. On the one hand, the first doped region and the second doped region affects other components in the semiconductor device; on the other hand, it also helps to further reduce the band-band tunneling effect inside the transistor, thereby further reducing the GIDL effect.

(57) The above are only the preferred embodiments of the present disclosure. It should be pointed out that for those skilled in the art, without departing from the principles of the present disclosure, several improvements and modifications can also be made, and these improvements and modifications should also be regarded as Scope of protection of this disclosure.

Claims

1. A semiconductor device comprising: a substrate; a plurality of transistors located on the substrate, wherein the plurality of transistors each comprises an active cylinder, wherein the active cylinder comprises a channel region, a source region and a drain region disposed on opposite sides of the channel region, a first doped region disposed between the source region and the channel regions, and a second doped region disposed between the drain region and the channel region; wherein the first doped region, the source region, the second doped region and the drain region comprise doping ions of a first type, and wherein a doped concentration of the first doped region is lower than a doped concentration of the source region, and a doped concentration of the second doped region is lower than a doped concentration of the drain region; and wherein each of the active cylinders extends along a first direction, wherein the active cylinders are arranged in an array along a second direction and a third direction, wherein the first direction and the second direction are both parallel to a top surface of the substrate, wherein the first direction intersects the second direction, and wherein the third direction is perpendicular to the top surface of the substrate direction; wherein the semiconductor device further comprises: a plurality of word lines extending along the second direction, wherein the plurality of word lines continuously covers the channel regions in the active cylinders spaced at intervals along the second direction; and wherein the plurality of word lines are arranged at the intervals along the third direction, and wherein among two adjacent ones of the plurality of word lines along the third direction, one of the two adjacent word lines closer to the substrate protrudes from the other of the two word lines along the third direction.

2. The semiconductor device of claim 1, wherein each of the active cylinders extends in a first direction, wherein the first direction is parallel to a top surface of the substrate; wherein the semiconductor device further comprises: a first isolation layer, located on the substrate, wherein the first isolation layer is disposed around an outer periphery of the first doped region, and wherein the active cylinder penetrates the first isolation layer along the first direction; and a second isolation layer, located on the substrate, wherein the second isolation layer is disposed around an outer periphery of the second doped region, and wherein the active cylinder penetrates the second isolation layer along the first direction.

3. The semiconductor device of claim 2, wherein the active cylinders of the plurality of transistors are spaced apart along a second direction, wherein the second direction is parallel to the top surface of the substrate, and wherein the first direction intersects the second direction; wherein the first isolation layer surrounds continuously the outer periphery of the first doped regions of the plurality of transistors arranged at intervals along the second direction; and wherein the second isolation layer surrounds continuously the outer periphery of the second doped regions of the plurality of transistors arranged at the intervals along the second direction.

4. The semiconductor device according to claim 3, further comprising: a first air gap, located between adjacent first doped regions; and a second air gap, located between adjacent second doped

regions.

5. The semiconductor device of claim 1, wherein the active cylinders of the plurality of transistors extend in the first direction, wherein the first direction is parallel to a top surface of the substrate; wherein the semiconductor device further comprises: a first isolation layer disposed on the substrate, wherein the first isolation layer covers the first doped region, wherein the active cylinders each penetrates the first isolation layer along the first direction; and a second isolation layer, disposed on the substrate, wherein the second isolation layer covers the second doped region, wherein the active cylinders each penetrates the second isolation layer along the first direction.

6. The semiconductor device of claim 5, wherein the active cylinders are spaced at intervals along a second direction, wherein the second direction is parallel to the top surface of the substrate, and wherein the first direction intersects the second direction; wherein the first isolation layer continuously covers each of the first doped regions spaced at the intervals along the second direction; and wherein the second isolation layer continuously covers each of the second doped regions spaced at the intervals along the second direction.

7. The semiconductor device of claim 1, wherein the channel region comprises doping ions of a first type; or, wherein the channel region comprises doping ions of a second type, wherein the doping ions of the first type and the doping ions of the second type have opposite conductivity.

8. The semiconductor device of claim 1, wherein each of the active cylinders extends in a first direction, wherein the first direction is parallel to a top surface of the substrate; wherein a length of the first doped region along the first direction is more than twice a thickness of each of the active cylinders along a third direction, wherein the third direction is perpendicular to the top surface of the substrate; and wherein a length of the second doped region along the first direction is more than twice a thickness of each of the active cylinders along the third direction.

9. The semiconductor device of claim 1, wherein each of the active cylinders extends in a first direction, wherein the first direction is parallel to a top surface of the substrate; wherein a length of the first doped region along the first direction is greater than or equal to a length of the source region along the first direction; and wherein a length of the second doped region along the first direction is greater than or equal to a length of the drain region along the first direction.

10. A method for forming a semiconductor device, comprising steps of: providing a substrate; forming a plurality of transistors on the substrate, wherein the plurality of transistors each comprises an active cylinder, wherein the active cylinder comprises a channel region, a source region and a drain region disposed on opposite sides of the channel region, a first doped region disposed between the source region and the channel region, and a second doped region disposed between the drain region and the channel region, wherein the first doped region, the source region, the second doped region, and the drain region all comprise doping ions of a first type, wherein a doped concentration of the first doped region is lower than a doped concentration of the source region, and wherein a doped concentration of the second doped region is lower than a doped concentration of the drain region; and wherein each of the active cylinders extends along a first direction, wherein the active cylinders are arranged in an array along a second direction and a third direction, wherein the first direction and the second direction are both parallel to a top surface of the substrate, wherein the first direction intersects the second direction, and wherein the third direction is perpendicular to the top surface of the substrate direction; wherein the semiconductor device further comprises: a plurality of word lines extending along the second direction, wherein the plurality of word lines continuously covers the channel regions in the active cylinders spaced at intervals along the second direction; and wherein the plurality of word lines are arranged at the intervals along the third direction, and wherein among two adjacent ones of the plurality of word lines along the third direction, one of the two adjacent word lines closer to the substrate protrudes from the other of the two word lines along the third direction.

11. The method for forming the semiconductor device according to claim 10, wherein the step of forming the plurality of transistors on the substrate comprises: forming a semiconductor cylinder

over the substrate; defining an initial channel region, an initial source region and an initial drain region disposed on opposite sides of the initial channel region, an initial first doped region between the initial source region and the initial channel region, and an initial second doped region between the initial drain region and the initial channel region; and implanting first type doping ions into the initial source region, the initial drain region, the initial first doped region and the initial second doped region to form the source region, the drain region, the first doped region, and the second doped region.

12. The method for forming the semiconductor device according to claim 11, wherein the step of defining the initial channel region, the initial source region and the initial drain region disposed on the opposite sides of the initial channel region in the semiconductor cylinder, and the initial first doped region located between the initial source region and the initial channel region, and the initial second doped region located between the initial drain region and the initial channel region, comprises steps of: defining the initial channel region in the semiconductor cylinder, disposing the initial source region and the initial drain region on opposite sides of the initial channel region along the first direction; defining the initial first doped region between the initial channel region and the initial source region along the first direction; and defining the initial second doped region between the initial drain region and the initial channel region along the first direction, wherein a length of the initial first doped region along the first direction is more than twice a thickness of the semiconductor cylinder along a third direction, wherein a length of the initial second doped region along the first direction is more than twice a thickness of the semiconductor cylinder in the third direction, and wherein the first direction is parallel to a top surface of the substrate, and wherein the third direction is perpendicular to the top surface of the substrate.

13. The method for forming the semiconductor device according to claim 11, wherein the step of forming the semiconductor cylinder on the substrate comprises: forming a stacked layer on the substrate, wherein the stacked layer comprises a semiconductor layer and a sacrificial layer alternately stacked along a third direction, wherein the third direction is perpendicular to the top surface of the substrate; etching the stacked layer to form a plurality of first trenches penetrating the stacked layer along the third direction, wherein the plurality of first trenches separates the semiconductor layer into a plurality of the semiconductor cylinders spaced at intervals along a second direction, wherein the plurality of the semiconductor cylinders extend along the first direction, wherein both the first direction and the second direction are parallel to the top surface of the substrate, and wherein the first direction intersects the second direction.

14. The method for forming the semiconductor device according to claim 13, wherein before the first type doping ions are implanted into the initial source region, the initial drain region, the initial first doped region and the initial second doped region, the method further comprises following steps: removing a part of the sacrificial layer in the stacked layer to form second trenches and third trenches, wherein the second trenches expose the semiconductor cylinder between the initial first doped region and the initial channel region and the semiconductor cylinder between the initial first doped region and the initial source region; wherein the third trenches expose the semiconductor cylinder between the initial second doped region and the initial channel region and the semiconductor cylinder between the initial second doped region and the initial drain region; and deposited a first dielectric material in the second trenches and in the third trenches; forming a first isolation layer in the second trenches; and forming a second isolation layer in the third trenches simultaneously.

15. The method of forming the semiconductor device according to claim 14, wherein before implanting the first type doping ions into the initial source region, the initial drain region, the initial first doped region, and the initial second doped region, the method further comprises: removing the sacrificial layer from the stacked layer to expose fourth trenches of the semiconductor cylinder; and filling a second dielectric material into the fourth trenches to form an interlayer isolation layer.

16. The method of forming the semiconductor device according to claim 15, wherein implanting

the first type doping ions into the initial source region, the initial drain region, the initial first doped region and the initial second doped region comprise steps of: implanting the first type doping ions into the initial source region and the initial drain region to form the source region and the drain region; and implanting the first type doping ions into the initial first doped region and the initial second doped region to form the first doped region and the second doped region.

17. The method for forming the semiconductor device according to claim 16, wherein after forming the source region, the first doped region, the drain region and the second doped region, the method further comprises following steps: removing the interlayer isolation layer between adjacent first doped regions and simultaneously removing the interlayer isolation layer between adjacent second doped regions to form a first air gap between the first doped regions and a second air gap between adjacent second doped regions simultaneously.

18. The method for forming the semiconductor device according to claim 17, wherein after forming the first air gap between adjacent first doped regions and the second air gap between adjacent second doped regions simultaneously, the method further comprises a step of: depositing a first dielectric material in the first air gap and the second air gap.

19. The method for forming the semiconductor device according to claim 11, further comprising: implanting the first type doping ions into the initial channel region to form the channel region; or, implanting the second type doping ions into the initial channel region to form the channel region, wherein the first type doping ions and the second type doping ions have opposite conductivity.
